MD IRSAD

Generative AI / AI Automation Engineer | Python & Cloud Data Engineer

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PROFESSIONAL SUMMARY

Python and Data Engineer with 3+ years of experience delivering backend APIs, serverless AWS workflows, ETL pipelines, and automation. Applies **LLM APIs, prompt automation, RAG concepts, REST integration, and Python data preparation** to practical AI-enabled workflows. Hands-on with **Python, SQL, Pandas, NumPy, FastAPI, Flask, AWS Lambda, S3, SQS, DynamoDB, Glue, PySpark, and MySQL**. Focused on GenAI application engineering and integrations; not presenting unverified LangChain, LangGraph, or machine-learning model training as production experience.

Core Expertise: Python | LLM API Integration | RAG Concepts | Prompt Automation | GenAI Applications | REST APIs | FastAPI | Data Pipelines | AWS Serverless | SQL | Pandas | PySpark | Workflow Automation

TECHNICAL SKILLS

Programming & Backend: Python 3.12/3.13 | SQL | FastAPI | Flask | REST APIs | SQLAlchemy | Pydantic | Boto3

Data Engineering & Analytics: PySpark | Pandas | NumPy | AWS Glue | ETL Pipelines | Data Transformation | Schema Mapping | Incremental Processing | Data Validation | Data Pre-processing

AWS & Cloud: S3 | Lambda | Glue | Step Functions | DynamoDB | Redshift | Athena | CloudWatch | SQS/SNS | IAM | API Gateway | ECR | ECS | Cognito

Databases & DevOps: MySQL | SQL | DynamoDB | Redshift | Docker | Git | GitHub | Poetry | Conda | CI/CD | Pytest | Unittest | Testcontainers | SonarQube

AI & Automation: LLM APIs | Prompt Automation | RAG Concepts | AI-assisted Workflows | Python Automation | Data Preparation for ML | Workflow Automation

Frontend & Tools: React.js | Next.js | Node.js | Express.js | Tailwind CSS | JavaScript/ES6+ | Axios | Selenium | Jupyter Notebook | Postman | DBeaver | VS Code | PyCharm

PROFESSIONAL EXPERIENCE

Data Engineer | Freelance – Client: HCLTech / MOTOR

Feb 2025 – Present

Finance Data Engineering & Cloud Automation

Work on Python and AWS-based finance data workflows, ETL pipelines, serverless processing, API integrations, document automation, monitoring, and production issue handling across client projects.

Key Contributions:

- Developed AWS Glue/PySpark ETL pipelines for invoice, remittance, and credit-memo data processing.
- Orchestrated multi-step workflows using Step Functions, Lambda, S3, DynamoDB, and Redshift.
- Implemented schema mapping, incremental processing, data validation, enrichment, and large-scale ingestion workflows.
- Integrated Oracle ORDS/REST APIs and automated finance-document generation with monitoring and error handling.

Tech Stack: Python | PySpark | Pandas | AWS Glue | Lambda | S3 | SQS | Step Functions | DynamoDB | Redshift | CloudWatch | Oracle ORDS | REST APIs

Software Developer | 33 Kuber India Advisory LLP

Sep 2023 – Jan 2025

Business Applications – Backend, APIs, Data & Automation

Developed Python backend services and full-stack business applications with REST APIs, MySQL, React, authentication, reporting, workflow modules, and database optimization.

Key Contributions:

- Built and maintained Python REST APIs using Flask and MySQL for business applications.
- Developed authentication, OTP login, password reset, CRM/CRS/SRC integration, task, attendance, timesheet, and leave workflows.
- Optimized MySQL using stored procedures, views, functions, indexing, and query improvements.
- Built React/Tailwind interfaces, integrated APIs, and supported application testing and production issue resolution.

Tech Stack: Python | Flask | FastAPI | MySQL | REST APIs | React.js | Tailwind CSS | SQL | Git | Pytest

PROJECTS

PRINTSOFT REPLACEMENT – Serverless Finance Document Automation

Client: AUS Logistics | **Period:** May 2025 – Present | **Role:** Data Engineer

Migrated a legacy Printsoft document workflow to a serverless AWS architecture for automated finance-document processing and storage.

- Built Lambda/SQS/S3 workflows for invoices, bills, statements, and credit memos.
- Integrated Oracle ORDS APIs and DynamoDB enrichment for automated document processing.

- Implemented conditional PDF/CSV templates, naming conventions, reprint logic, and output storage.
- Reduced manual document-processing effort by 80% through workflow automation and audit-friendly storage.

Tech Stack: Python | AWS Lambda | SQS | S3 | DynamoDB | Oracle ORDS | REST APIs | PDF/CSV Automation

PROD FINANCE DATA PIPELINE – AWS Data Engineering

Client: MOTOR | **Period:** Feb 2025 – May 2025 | **Role:** Data Engineer

Designed and deployed an AWS finance-data pipeline for invoice, remittance, and credit-memo processing using distributed ETL and workflow orchestration.

- Developed AWS Glue/PySpark ETL processing for approximately 30GB+ of daily data.
- Implemented schema mapping, incremental loads, bookmarks, transformation, and data-quality workflows.
- Orchestrated processing with Step Functions and Lambda, using S3 as the data-lake layer.
- Added CloudWatch monitoring, SES alerts, Teams notifications, and error handling; reduced downtime by 40%.

Tech Stack: Python | PySpark | AWS Glue | S3 | Lambda | Step Functions | DynamoDB | Redshift | CloudWatch

JAZZTRACK MANAGEMENT SYSTEM – Business Application Platform

Company: 33 Kuber India Advisory LLP | **Period:** Sep 2023 – Jan 2025 | **Role:** Software Developer

Built full-stack business applications with Python APIs, React, MySQL, authentication, workflow modules, and integrated business systems.

- Developed Flask/REST APIs and React/Tailwind modules for business workflows and reporting.
- Integrated CRM, CRS, and SRC systems to support centralized customer and operational data.
- Optimized MySQL using stored procedures, views, functions, and database-level improvements.
- Built OTP authentication, task tracking, attendance, project, timesheet, and leave-management workflows.

Tech Stack: Python | Flask | MySQL | REST APIs | React.js | Tailwind CSS | SQL | Git

GENAI / AUTOMATION APPLICATIONS

AI/Automation Focus: Applies Python automation, LLM APIs, prompts, and RAG concepts to workflow applications. Strongest fit: **Generative AI Engineer** roles emphasizing Python, API integration, RAG application fundamentals, cloud automation, and data pipelines rather than research-heavy ML model development.

Transferable AI Engineering Skills: Python | LLM APIs | Prompt Engineering/Automation | RAG Concepts | REST APIs | Data Processing | Pandas | NumPy | SQL | AWS | Workflow Automation

EDUCATION

B.Sc. Information Technology — University of Mumbai | Sree Narayana Guru College of Commerce | Graduated 2023 | 9.13 CGPA

Class XII — R.R.M.Y Inter College High School+2 | Board of Intermediate Education | 2020 | 73.2%

Class X — S.R.P.N. High School | Board of Secondary Education | 2018 | 63%

TARGET ROLES

Target Roles: Generative AI Engineer | AI Automation Engineer | Python Backend Engineer | Data Engineer | Cloud Data Engineer